

Summary:

MSc graduate in Applied Artificial Intelligence and Data Analytics with hands-on experience designing, developing, and deploying end-to-end machine learning and deep learning solutions. Proficient in Python, SQL, scikit-learn, TensorFlow, and FastAPI, with experience building production-ready ML pipelines, data preprocessing workflows, model evaluation, and cloud-deployed AI applications. Contributed to open-source machine learning projects and developed scalable AI solutions across predictive analytics, medical imaging, and retrieval-augmented generation (RAG) systems. Passionate about applying data-driven approaches to solve real-world problems and thrive in collaborative, fast-paced environments.

EDUCATION:

University of Bradford, Bradford, UK	Jan 2024 – Mar 2025
<i>MSc in Applied Artificial Intelligence and Data Analytics</i>	
Relevant Courseworks: Machine Learning, Business Data Analytics, Artificial Intelligence, Big Data Visualization	
University of Science and Technology Chittagong, Chattogram, Bangladesh	Jan 2015 – April 2019
<i>BSc Engineering in Electrical and Electronic Engineering</i>	

SKILLS:

- **Programming languages:** Python, SQL, R, Python, MATLAB
- **ML/AI:** Scikit-learn, PyTorch, TensorFlow, Explainable AI (Grad-CAM, SHAP), LLMs, RAG, Hybrid Retrieval, Vector
- Databases, Predictive Modelling
- **Data & Visualisation:** Pandas, NumPy, Matplotlib, Power BI, Excel
- **Tools:** FastAPI, Streamlit, Docker Compose, GitHub Actions (CI/CD), pytest
- **Databases:** MySQL, ChromaDB, Supabase
- **Frontend:** Next.js
- **Soft Skills:** Problem-solving, Collaboration, Communication, Agile Mindset

PROJECTS:

Kidney Tumor Identification System – [TensorFlow FastAPI Streamlit Docker MLflow DVC] [GitHub Repo]
• Developed an end-to-end deep learning system to classify kidney CT scans into four categories (Normal, Cyst, Tumor, and Stone) using an EfficientNetB4-based model, achieving 99.46% validation accuracy and 98.38% test AUC-ROC. • Built a production-ready application with FastAPI and Streamlit, incorporating Grad-CAM visualizations, uncertainty estimation, PDF report generation, and containerized deployment using Docker. • Applied MLOps practices including MLflow experiment tracking, DVC for data versioning, GitHub Actions for CI/CD, and automated data drift monitoring to support a reproducible machine learning workflow.
MediQuery – Role-Gated Medical RAG System – [FastAPI Next.js ChromaDB BM25 Docker Groq LLM] [GitHub Repo]
• Developed a role-based medical knowledge assistant using Retrieval-Augmented Generation (RAG), combining BM25 and vector search with cross-encoder reranking to deliver context-aware, cited responses. • Designed a secure retrieval pipeline with JWT-based role-based access control (RBAC), ensuring users could only access documents authorized for their roles before retrieval was performed. • Built and containerized the full-stack application with FastAPI, Next.js, ChromaDB, and Docker, integrating conversation-aware query rewriting and query safety checks for reliable AI-assisted knowledge retrieval.
London Bike-Sharing Demand Prediction – [Python scikit-learn Streamlit] [GitHub Repo]
• Developed an end-to-end machine learning pipeline to predict hourly bike-sharing demand using environmental and temporal data, achieving an R² score of 0.957 with a Random Forest model after pipeline optimization. • Performed data preprocessing, feature engineering, model comparison, and evaluation to identify the most effective regression model for demand forecasting. • Deployed the solution as an interactive Streamlit application with modular architecture, centralized YAML configuration, and an automated model training workflow for seamless cloud deployment.

CERTIFICATIONS / AWARDS:

• Python: Beginner to Advanced For Data Professionals	Apr 2025
• Get Job Ready: Power BI Data Analytics for All Levels	Oct 2023
• SQL Beginner to Advanced For Data Professionals	Dec 2023
• Complete the Google Data Analytics Professional Certification	Apr 2023

PROFESSIONAL DEVELOPMENT:

• Information Visualization	July 2024
<i>Aarhus University (via University of Bradford) Funded by the Turning Scheme UK</i>	
• Making Sense of Data: Handling, Processing, Visualizing, and Analyzing Data in R	Aug 2024
<i>Aarhus University (via University of Bradford) Funded by the Turning Scheme UK</i>	